BOUNDED COHOMOLOGY OF MEASURE-PRESERVING HOMEOMORPHISM GROUPS OF NON-ORIENTABLE SURFACES

MICHAEL BRANDENBURSKY AND LIOR MENASHE

Abstract. Let N_g be a closed non-orientable surface of genus $g \geq 3$, and let $\mathrm{Homeo}_0(N_g, \mu)$ be the identity component of its group of measure-preserving homeomorphisms. For every $n \geq 2$ we construct a linear injection

$$\overline{EH}_b^n(F_2) \hookrightarrow \overline{EH}_b^n\big(\mathrm{Homeo}_0(N_g, \mu)\big),$$

where $\overline{EH}_b^n$ denotes reduced exact bounded cohomology. In particular, $H_b^2(\mathrm{Homeo}_0(N_g, \mu))$ and $H_b^3(\mathrm{Homeo}_0(N_g, \mu))$ are infinite-dimensional. The main new issue is low genus. A natural figure-eight subgroup $F_2 \leq \pi_1(N_g)$ is hyperbolically embedded for $g \geq 4$, whereas in genus three hyperbolic embeddedness fails for every π_1-injective figure-eight with orientable regular neighborhood. We overcome this obstruction using the amalgam decomposition $\pi_1(N_g) = F_2 *_{\mathbb{Z}} F_{g-2}$ and an isometric bounded-cohomology extension theorem for graphs of groups with amenable edge groups. Combined with the Gambaudo–Ghys transfer and annular pushing, this yields the injection above.

1. Introduction

Gromov's seminal work [15] established bounded cohomology as a fundamental tool in geometry and topology, with applications to the geometry of manifolds, stable commutator length, amenable groups, and symplectic geometry. In low degrees it is understood for many classes of finitely generated groups. In higher degrees, however, even the bounded cohomology of the non-abelian free group F_2 remains much less understood; see [13].

Since the early 2000s, bounded cohomology of transformation groups has received considerable attention. Degree-two bounded classes are closely related to quasimorphisms, which detect geometric and dynamical phenomena such as entropy and symplectic spectral invariants; see [3, 19]. Higher-degree bounded classes of transformation groups were studied more recently by Brandenbursky–Marcinkowski [4], Kimura [20], and Nitsche [23]. In the orientable case, the Gambaudo–Ghys–Polterovich transfer method provides a powerful mechanism for producing such classes. The non-orientable case has

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a genuinely different low-genus feature: the natural geometric free subgroup needed by the transfer argument is hyperbolically embedded for $g \geq 4$, but in genus three it necessarily fails even the almost malnormality condition. Resolving this obstruction is one of the main points of the present paper.

Let N_g be a closed non-orientable surface of genus $g \geq 3$. Fix a hyperbolic metric on N_g and let μ be its Riemannian area measure, normalized to have total area one. Equivalently, the lift of μ to the orientation double cover $\Sigma_{g-1} \to N_g$ is induced by the lifted hyperbolic area form. Let $\mathrm{Homeo}(N_g, \mu)$ be the group of μ-preserving homeomorphisms, equipped with the compact-open topology, and let $\mathrm{Homeo}_0(N_g, \mu)$ denote its identity component. Fathi's local-contractibility theorem [11] implies that this identity component is path connected. In particular, every element of $\mathrm{Homeo}_0(N_g, \mu)$ is joined to the identity by an isotopy through μ-preserving homeomorphisms. Our main result is the following.

Theorem 1. *Let $g \geq 3$. Then*

$$\dim H_b^2(\mathrm{Homeo}_0(N_g, \mu)) = \dim H_b^3(\mathrm{Homeo}_0(N_g, \mu)) = \infty.$$

For every $g \geq 3$ and every $n \geq 2$ we construct a linear injection

$$\overline{EH}_b^n(F_2) \hookrightarrow \overline{EH}_b^n\big(\mathrm{Homeo}_0(N_g, \mu)\big). \tag{1}$$

Since $\overline{EH}_b^2(F_2)$ and $\overline{EH}_b^3(F_2)$ have dimension $2^{\aleph_0}$, the theorem follows; in fact, in these two degrees the targets contain subspaces of dimension $2^{\aleph_0}$.

We now describe the main ingredients. Choose a figure-eight $i : S^1 \vee S^1 \hookrightarrow N_g$ contained in an orientable once-holed torus. Writing $G_g = \pi_1(N_g)$ and identifying the image of i_* with $H = F_2$, van Kampen gives an amalgam decomposition $G_g = F_2 *_{\mathbb{Z}} F_{g-2}$. For $g \geq 4$ we prove directly that this geometric subgroup H is hyperbolically embedded in G_g with respect to a natural finite relative generating set. The proof includes an explicit computation of the relative metric. We also give an independent proof using Bowditch's relative hyperbolicity criterion. The extension theorem of Frigerio–Pozzetti–Sisto [12] then gives surjectivity of restriction from the reduced exact bounded cohomology of G_g to that of F_2.

Genus three is qualitatively different. In the presentation

$$G_3 = \langle a, b, c \mid [a, b]c^2 = 1 \rangle, \qquad H = \langle a, b \rangle,$$

one has $\langle c^2 \rangle \subseteq H \cap cHc^{-1}$, so H is not almost malnormal and hence cannot be hyperbolically embedded. We prove a geometric version of this obstruction: no π_1-injective embedded figure-eight in N_3 with orientable regular neighborhood has almost malnormal image. Rather than replacing the figure-eight by a less geometric subgroup, we use the cyclic amalgam itself. The edge group is amenable, and the isometric extension theorem of Bucher–Burger–Frigerio–Iozzi–Pagliantini–Pozzetti [8] produces, for every $g \geq 3$ and $n \geq 2$, an

isometric linear right inverse

$$J_g^n : H_b^n(F_2) \longrightarrow H_b^n(G_g)$$

to restriction. This works uniformly in all genera and is stronger than the surjectivity required for the proof.

Finally, we combine this cohomological section with the Gambaudo–Ghys–Polterovich transfer. The figure-eight lies in an orientable once-holed torus, so the annular pushing construction can be carried out locally inside the non-orientable surface. We prove a norm approximation showing that, after pullback by the pushing homomorphisms, the transferred class converges to a positive multiple of its restriction to F_2. Composing the transfer with the isometric section therefore gives the injection (1). This separates the two genuinely different issues in the proof: the geometric behavior of the figure-eight subgroup and the bounded-cohomological extension across the cyclic amalgam.

Remark 1.1. Kumar in his thesis [21] constructed an infinite-dimensional space of homogeneous quasimorphisms on $\mathrm{Homeo}_0(N_g, \mu)$. Our argument gives a direct bounded-cohomological proof of the infinite-dimensionality in degree two as well as degree three, and the construction (1) applies in every degree $n \geq 2$.

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2. Preliminaries

All cohomology groups in this paper have trivial real coefficients, which we omit from the notation. Transformation groups are regarded as discrete groups when taking their bounded cohomology. For $g \geq 3$ we write

$$G_g = \pi_1(N_g, z), \qquad \mathcal{H}_g = \mathrm{Homeo}_0(N_g, \mu).$$

For a homomorphism f, we denote by f^* its pullback in bounded cohomology, and by f^n the induced map in degree n on reduced exact bounded cohomology. These pullbacks are seminorm non-increasing on bounded cohomology and exact bounded cohomology, and norm non-increasing on their reduced versions. Indeed, the cochain pullback is given by precomposition and does not increase the supremum norm; the inequality passes to cohomology and to the quotient by zero-seminorm classes.

2.1. **Bounded cohomology.** Let G be a group and let $C_b^n(G)$ be the space of bounded functions $G^{n+1} \to \mathbb{R}$. We use the left homogeneous convention: the action is $(g \cdot f)(g_0, \dots, g_n) = f(g^{-1}g_0, \dots, g^{-1}g_n)$, and $C_b^n(G)^G$ denotes the invariant subspace. The coboundary is

$$\delta^n f(g_0, \dots, g_{n+1}) = \sum_{j=0}^{n+1} (-1)^j f(g_0, \dots, \widehat{g_j}, \dots, g_{n+1}).$$

The cohomology of this homogeneous cochain complex is $H_b^n(G)$. Equivalently, one can use the inhomogeneous cochain complex $\bar{C}_b^n(G) = \ell^\infty(G^n)$, with G^0 a singleton. The isometric identification with homogeneous cochains is given by

$$\phi(f)(g_1, \dots, g_n) = f(1, g_1, g_1g_2, \dots, g_1 \cdots g_n),$$

and the inhomogeneous differential is $\bar{\delta} = \phi\delta\phi^{-1}$. We refer to [13] for these conventions and the basic properties of bounded cohomology.

2.1.1. *Exact and reduced bounded cohomology.* The exact bounded cohomology is

$$EH_b^n(G) = \ker\big(c_G^n : H_b^n(G) \longrightarrow H^n(G)\big),$$

where c_G^n is the comparison map. The supremum norm on cochains induces

$$\|\xi\| = \inf\{\|c\|_\infty : [c] = \xi\}$$

on $H_b^n(G)$. Write $N^n(G)$ for its subspace of zero-seminorm classes. We use the notation

$$\overline{H}_b^n(G) = H_b^n(G)/N^n(G), \qquad \overline{EH}_b^n(G) = EH_b^n(G)/(EH_b^n(G) \cap N^n(G)).$$

Both reduced spaces carry genuine norms. Naturality of the comparison map and the norm inequality for pullbacks imply that every group homomorphism induces maps on these spaces.

2.2. **Quasimorphisms.** A map $f : G \to \mathbb{R}$ is a *quasimorphism* if

$$D(f) := \sup_{g,h \in G} |f(gh) - f(g) - f(h)| < \infty.$$

It is *homogeneous* if $f(g^k) = kf(g)$ for all $g \in G$ and $k \in \mathbb{Z}$. Let $Q(G, \mathbb{R})$ and $Q^h(G, \mathbb{R})$ denote the corresponding vector spaces. The map $f \mapsto [\bar{\delta}f]$ gives

$$EH_b^2(G) \cong \frac{Q(G, \mathbb{R})}{\bar{C}_b^1(G) + \mathrm{Hom}(G, \mathbb{R})} \cong \frac{Q^h(G, \mathbb{R})}{\mathrm{Hom}(G, \mathbb{R})};$$

see [13, Chapter 2].

2.3. **Hyperbolically embedded subgroups.** We recall the definition from [9, Section 4]. Cayley graphs have edges of length one, and paths may traverse generator edges in either direction. Let $H \leq G$ and $X \subseteq G$ with $G = \langle X \cup H \rangle$. The alphabet $X \sqcup H$ is a disjoint union, so labels from its two parts are distinguished even when they represent the same group element. A path in $\Gamma(G, X \sqcup H)$ is *admissible* if it uses no edge of the distinguished subgraph $\Gamma(H, H)$. In particular, H-labeled edges in other left cosets of H are allowed. For $h, k \in H$, define $\hat{d}(h, k)$ as the minimum length of an admissible path from h to k, with value ∞ when no such path exists. This is an extended metric on H.

Definition 2.1. We write $H \hookrightarrow_h (G, X)$ if:

(a) $G = \langle X \cup H \rangle$;

(b) $\Gamma(G, X \sqcup H)$ is hyperbolic;

(c) every ball of finite radius in $(H, \hat{d})$ is finite.

We write $H \hookrightarrow_h G$ if such an X exists.

We will verify these conditions directly. The following edge-addition criterion will be used for condition (b).

Proposition 2.2 ([24, Lemma 5.5]; see also [17, Corollary 2.4])**.** Suppose that a graph Γ_H is obtained from a hyperbolic graph Γ by adding edges. If there exists $M > 0$ such that every Γ-geodesic joining the endpoints of an edge of Γ_H has diameter at most M in Γ_H, then Γ_H is hyperbolic.

Definition 2.3. A subgroup $H \leq G$ is *malnormal* if $H \cap xHx^{-1} = \{1\}$ for every $x \in G \setminus H$. It is *almost malnormal* if all these intersections are finite.

Proposition 2.4 ([9, Proposition 4.33])**.** A hyperbolically embedded subgroup is almost malnormal.

2.3.1. *Examples.*

- The subgroup $\langle a^2, ab, ab^{-1} \rangle \leq F(a, b)$ is not hyperbolically embedded. It is the index-two kernel of the map sending both a and b to the nontrivial element of $\mathbb{Z}/2\mathbb{Z}$, and hence is not almost malnormal.
- The subgroup $\langle a^n b a^{-n} : n \geq 0 \rangle \leq F(a, b)$ has infinite intersection with its conjugate by a, although a does not belong to it. It is therefore not hyperbolically embedded.
- For $k > 2$, the free factor $F(a_1, a_2)$ is hyperbolically embedded in $F(a_1, \ldots, a_k)$ relative to $X = \{a_3, \ldots, a_k\}$. The ordinary geodesic between vertices of a left coset of this free factor stays in that coset, so Proposition 2.2 applies. Normal forms for the free product show that $\hat{d}(1, h) = \infty$ for $h \neq 1$, which proves condition (c).

For the geometric figure-eight embedding used below, hyperbolic embeddedness holds for $g \geq 4$, but fails for $g = 3$. In genus three we instead use an isometric bounded-cohomology extension over an amenable amalgam; see Proposition 4.10.

3. Gambaudo–Ghys map

We recall the transfer construction of [4], extending the constructions of Gambaudo–Ghys [14] and Polterovich [25], with conventions compatible with our left homogeneous cochains. Fix $g \geq 3$. Outside the cut locus of z, let $\alpha_{z,x}$ be the unique minimizing geodesic from z to x; on the cut locus, which has measure zero, choose the paths arbitrarily. This gives a measurable family of reference paths, as in [4, Section 2]. Put $\alpha_{x,z} = \alpha_{z,x}^{-1}$. For $f \in \mathcal{H}_g$, choose an isotopy $\{f_t\}_{0 \leq t \leq 1}$ from the identity to f and set

$$\gamma(f, x) = [\alpha_{z,x} * \{f_t(x)\}_{t \in [0,1]} * \alpha_{f(x),z}] \in G_g.$$

Here $*$ denotes traversal from left to right, while multiplication of based loop classes is taken in composition order: $[q][p]$ means first p, then q. With this convention the trajectory cocycle has the identity stated in Proposition 3.2 below.

Proposition 3.1. The element $\gamma(f, x)$ is independent of the chosen isotopy.

Proof. For a loop $\{u_t\}$ in $\mathcal{H}_g$ and a based loop δ in N_g, the square $(s, t) \mapsto u_t(\delta(s))$ shows that the evaluation loop $t \mapsto u_t(z)$ commutes with $[\delta]$. Thus the image of

$$(ev_z)_* : \pi_1(\mathcal{H}_g, \mathrm{Id}) \longrightarrow G_g$$

is contained in $Z(G_g)$. This center is trivial: a central deck transformation of $\mathbb{H}^2$ would commute with two hyperbolic elements with distinct pairs of fixed points at infinity, and consequently fix at least three boundary points. It must therefore be the identity. Such elements exist because G_g is a cocompact, non-elementary surface group; see [6, 10]. The same argument applies at any basepoint x. Concatenating one isotopy with the reverse of another therefore gives a null-homotopic evaluation loop, proving the claim. □

Proposition 3.2. For $f, h \in \mathcal{H}_g$ and $x \in N_g$,

$$\gamma(fh, x) = \gamma(f, h(x))\gamma(h, x).$$

Proof. Use the isotopy that first follows h_t and then $f_t \circ h$. The trajectory of x is the trajectory of $h_t(x)$ followed by that of $f_t(h(x))$. The two intermediate reference paths cancel, and the formula follows from the multiplication convention. □

To use left homogeneous cochains, define

$$\eta(f, x) = \gamma(f^{-1}, x)^{-1}. \tag{2}$$

Proposition 3.2 gives

$$\eta(hf,x)=\eta(h,x)\eta(f,h^{-1}(x)). \tag{3}$$

For $c\in C_b^n(G_g)^{G_g}$ put

$$I_b^n(c)(f_0,\dots,f_n)=\int_{N_g} c\big(\eta(f_0,x),\dots,\eta(f_n,x)\big)\,d\mu(x). \tag{4}$$

The trajectory cocycle is measurable; see [4, Section 2] and [5, Section 3.B]. Thus the integral is well defined.

Proposition 3.3. The maps I_b^n take homogeneous cochains to homogeneous cochains, commute with the coboundary, and satisfy

$$\|I_b^n(c)\|_\infty\le\mu(N_g)\|c\|_\infty.$$

Proof. The norm estimate is immediate, and the alternating sum defining the coboundary commutes with integration. For homogeneity, use (3), left invariance of c, and invariance of μ to obtain

$$\begin{aligned} I_b^n(c)(hf_0,\dots,hf_n)&=\int_{N_g} c\big(\eta(f_0,h^{-1}(x)),\dots,\eta(f_n,h^{-1}(x))\big)\,d\mu(x)\\ &=I_b^n(c)(f_0,\dots,f_n). \end{aligned}$$

□

It follows that I_b^n induces $\Gamma_b^n:H_b^n(G_g)\to H_b^n(\mathcal{H}_g)$. It also preserves exact classes. Indeed, for a fixed f, the map $x\mapsto\gamma(f,x)$ has essentially finite image, as in [4, Lemma 2.2]. Consequently (4) is also defined on arbitrary homogeneous cochains. The same calculation as in Proposition 3.3 therefore gives a cochain map on ordinary homogeneous cochains, and the resulting square with the comparison maps commutes. The norm estimate then gives the induced map

$$\overline{E\Gamma}_b^n:\overline{EH}_b^n(G_g)\longrightarrow\overline{EH}_b^n(\mathcal{H}_g).$$

4. Proofs

The reduced exact bounded cohomology of F_2 has dimension $2^{\aleph_0}$ in degrees two and three; for degree two see [7] and [13, Section 2.6], and for degree three see [27] and [12, Corollary 3]. We will find an embedded figure-eight $i:S^1\vee S^1\hookrightarrow N_g$, contained in an orientable once-holed torus, for which

$$i^n:\overline{EH}_b^n(G_g)\longrightarrow\overline{EH}_b^n(F_2)$$

is surjective for every $n\ge 2$. For $g\ge 4$ we first prove this using a direct hyperbolic-embedding argument. For $g=3$, where that geometric subgroup is not hyperbolically embedded, we construct an isometric linear right inverse to restriction using an amenable amalgam. The latter construction in fact works for every $g\ge 3$.

As in [4], annular pushing maps then give homomorphisms $\rho_\epsilon : F_2 \to \mathcal{H}_g$ and a constant $\Lambda > 0$ such that

$$\left\|\rho_\epsilon^n \overline{E\Gamma}_b^n(\xi) - \Lambda i^n(\xi)\right\| \longrightarrow 0 \quad \text{as } \epsilon \to 0$$

for every $\xi \in \overline{EH}_b^n(G_g)$. Thus the diagram

$$\begin{array}{ccc} \overline{EH}_b^n(G_g) & \xrightarrow{\overline{E\Gamma}_b^n} & \overline{EH}_b^n(\mathcal{H}_g) \\ \downarrow{\scriptstyle i^n} & \swarrow{\scriptstyle \rho_\epsilon^n} & \\ \overline{EH}_b^n(F_2) & & \end{array}$$

commutes up to multiplication by Λ and an error tending to zero. This will prove

$$\dim H_b^n(\mathcal{H}_g) \geq \dim \overline{EH}_b^n(F_2), \qquad n \geq 2.$$

4.1. **An elementary retraction when $g \geq 5$.** We retain the elementary argument in higher genus, but use j and r to distinguish this figure-eight from the figure-eight embedding i used later.

Proposition 4.1. For $g \geq 5$ there are an embedding $j : S^1 \vee S^1 \hookrightarrow N_g$ and a retraction $r : N_g \to S^1 \vee S^1$ such that $r \circ j = \mathrm{Id}$.

Proof. Write $N_g = \Sigma_2 \# N_{g-4}$. Choose an embedded figure-eight given by one meridian in each of the two handles, joined at the common basepoint as in Figure 1. In the standard CW description the surface group has presentation

$$\left\langle x_1, y_1, x_2, y_2, d_1, \ldots, d_{g-4} \;\middle|\; [x_1, y_1][x_2, y_2] d_1^2 \cdots d_{g-4}^2 = 1 \right\rangle.$$

Define a map of the one-skeleton to the chosen figure-eight that is the identity on x_1, x_2 and collapses the remaining generator loops. The attaching loop of the two-cell maps to a null-homotopic loop, so by the usual CW extension criterion this map extends over the two-cell. It is the required retraction. Equivalently, its induced homomorphism sends x_1, x_2 to the two free generators and all other generators to 1. □

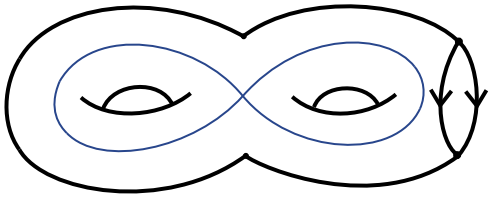


FIGURE 1. The embedding $j : S^1 \vee S^1 \hookrightarrow N_5$ and the retraction $r : N_5 \to S^1 \vee S^1$.

Corollary 4.2. For $n \geq 2$, $j^n \circ r^n = \mathrm{Id}$. In particular, j^n is surjective and r^n is an isometric injection.

Proof. Functoriality gives the identity. Since both pullbacks are norm non-increasing, $\|u\| = \|j^n r^n(u)\| \leq \|r^n(u)\| \leq \|u\|$. □

4.2. **The free subgroup and its amalgam decomposition.** For the rest of the proof, let $g \geq 3$ and put $m = g - 2$. Decompose

$$N_g = T \cup_\delta N_{m,1}, \qquad T \cong \Sigma_{1,1}, \quad \delta = \partial T = \partial N_{m,1}.$$

Here $\Sigma_{1,1}$ is a compact once-holed torus, and $N_{m,1}$ is a compact non-orientable surface of genus m with one boundary component. Choose meridian and longitude loops α, β meeting transversely only at z, whose union is a deformation retract of T. This defines $i : S^1 \vee S^1 \hookrightarrow N_g$ in the orientable part of $\mathbb{T}^2 \# N_m$; see Figure 2. Write a, b for their based homotopy classes and $w = [a, b] = aba^{-1}b^{-1}$.

With suitable choices of paths to the boundary and of generator orientations, van Kampen's theorem gives

$$\begin{aligned} G_g &= \langle a, b, c_1, \dots, c_m \mid [a, b] c_1^2 \cdots c_m^2 = 1 \rangle \\ &= H *_C B, \qquad H = F(a, b), \quad B = F(c_1, \dots, c_m), \end{aligned} \tag{5}$$

where, putting $v = c_1^2 \cdots c_m^2$, the abstract edge group and its injections are

$$C_0 = \langle t \rangle \cong \mathbb{Z}, \qquad H \xleftarrow{t \mapsto w} C_0 \xrightarrow{t \mapsto v^{-1}} B. \tag{6}$$

Both boundary words are nontrivial and of infinite order in their free groups, so the maps in (6) are injective. The normal-form theorem for amalgams [26, Chapter I, Section 1] therefore identifies H and B with subgroups of G_g, and

$$H \cap B = C = \langle w \rangle = \langle v \rangle, \qquad w = v^{-1}.$$

Thus $i_* : F_2 \to G_g$ is injective for every $g \geq 3$. We identify F_2 with H by this map and retain i^n for the induced restriction on reduced exact bounded cohomology.

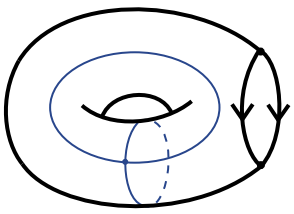


FIGURE 2. The figure-eight embedding $i : S^1 \vee S^1 \hookrightarrow N_g$, illustrated for $g = 3$. The complementary piece has $g - 2$ crosscaps.

4.3. **A direct hyperbolic embedding for $g \geq 4$.**

Proposition 4.3. Let $g \geq 4$, $m = g - 2$, and use the notation of (5). Then

$$H \hookrightarrow_h (G_g, X), \qquad X = \{c_1, \dots, c_m\}.$$

More precisely, for the relative metric defined by this finite set X,

$$\hat{d}(1, h) = \begin{cases} 2m|k|, & h = w^k, \quad k \in \mathbb{Z}, \\ \infty, & h \in H \setminus C. \end{cases} \tag{7}$$

Proof. We verify Definition 2.1 directly. Condition (a) is immediate. Write

$$\Gamma = \Gamma(G_g, \{a, b, c_1, \dots, c_m\}), \qquad \Gamma_H = \Gamma(G_g, X \sqcup H).$$

The graph Γ is hyperbolic because G_g acts properly and cocompactly on $\mathbb{H}^2$; see [6, Chapters I.8 and III.H]. We first establish the path estimates needed for (b) and (c).

Subpaths leaving and returning to a factor. Let $\mathcal{T}$ be the Bass–Serre tree of $H *_C B$. Its vertices are $G_g/H \sqcup G_g/B$ and its edges are G_g/C, with uC joining uH to uB; see [26, Chapter I, Section 4]. Map a group vertex u to the midpoint of uC. Map an H-labelled edge through the corresponding H-vertex, and a $c_j^{\pm 1}$-edge through the corresponding B-vertex. The same construction applies to Γ, with its a, b-edges mapped through H-vertices.

Consequently, a path that leaves B and returns to B, with endpoints $x, y \in B$ and all intermediate vertices outside B, has $xC = yC$: its image avoids the vertex B and must return along the same incident edge of the tree. The analogous statement holds for H.

Overlap of boundary axes. Let $T_B = \Gamma(B, X)$ be the ordinary Cayley tree of B, and let L be the axis of $v = c_1^2 \cdots c_m^2$, containing 1. Its positive labels repeat the word $c_1 c_1 c_2 c_2 \cdots c_m c_m$. Each oriented two-letter subword in this cyclic word occurs in exactly one position modulo its length $2m$: these subwords are $c_j c_j$ and $c_j c_{j+1}$, where $c_{m+1} = c_1$. Here we use $m \geq 2$. If L and uL share two consecutive edges, their positive orientations agree on the common segment, since all positive labels belong to X and none belongs to X^{-1}. The two-letter word then determines the same phase on both lines. Uniqueness of labelled paths in a Cayley tree forces the entire lines, with their phases, to agree. Their phase-zero vertices are respectively C and uC, so $u \in C$. Therefore

$$\operatorname{diam}_{T_B}(L \cap uL) \leq 1 \qquad (u \in B \setminus C). \tag{8}$$

If $p_L : T_B \to L$ is nearest-point projection, it follows that

$$\operatorname{diam}_{T_B} p_L(uC) \leq 1 \qquad (u \in B \setminus C). \tag{9}$$

Indeed, $uC \subset uL$; two disjoint lines in a tree project to a single point, and intersecting lines project onto their intersection. See Figure 3(b) for the case of a one-edge intersection.

A weighted normal-form observation. Give the a, b-edges of Γ length 1 and the c_j-edges length $2/m$, and write d_* for the resulting path metric. The intrinsic weighted lengths of the two representatives of every edge-group element agree:

$$|w^k|_{\{a,b\}} = 4|k| = \frac{2}{m}|v^{-k}|_X. \tag{10}$$

Both factors are isometrically embedded on vertices for these weighted metrics. We give the normal-form argument explicitly. Let W be a word in $\{a^{\pm 1}, b^{\pm 1}, c_1^{\pm 1}, \dots, c_m^{\pm 1}\}$ representing an element $h \in H$. Split W into maximal alternating H- and B-blocks and replace each block by a shortest word in its factor. If more than one block remains, the amalgam normal-form theorem implies that some block representing an element of C can be transferred to the adjacent factor; otherwise the alternating word would be reduced and could not represent an element of H. Transfer such a block using (10) and merge the adjacent blocks. This does not increase weighted length and decreases the number of blocks. Iterating leaves a single block. If that block lies in B, then, because it represents the element $h \in H$, it belongs to $H \cap B = C$ and can be transferred once more to H without changing weighted length. We have therefore produced an H-word of weighted length at most that of W. The reverse inequality is immediate because an intrinsic H-word is also an ambient word. The argument for an element of B is identical. Thus both factors are isometrically embedded. In particular,

$$d_*(1,h) = |h|_{\{a,b\}} \ (h \in H), \qquad d_*(x,y) = \frac{2}{m} d_{T_B}(x,y) \ (x,y \in B). \tag{11}$$

Condition (b): hyperbolicity of Γ_H. Suppose first that $m > 2$. If an ordinary Γ-geodesic q with endpoints $1, h \in H$ used a c_j-edge, its weighted length would be strictly smaller than its ordinary length. Since an H-word also joins these endpoints, (11) would imply

$$|h|_{\{a,b\}} = d_*(1,h) \le \ell_*(q) < \ell(q) = d_\Gamma(1,h) \le |h|_{\{a,b\}},$$

a contradiction. Hence every such geodesic stays in the ordinary H-Cayley tree. Its vertices have diameter at most 1 in Γ_H; including points in edge interiors gives a bound of 2.

It remains to consider $m = 2$, i.e. $g = 4$. Now all the weights are one, so (11) says that the ordinary factor Cayley trees are isometrically embedded on vertices in Γ. Let q be a subpath of a Γ-geodesic whose endpoints lie in H and whose intermediate vertices lie outside H. After left translation by its initial vertex, the preceding Bass–Serre-tree observation gives endpoints $1, v^k$, and no intermediate vertex of q lies in H.

We claim that q stays in B. Otherwise, consider each maximal subpath r of q whose endpoints x, y lie in B and whose intermediate vertices lie outside B, and replace r by the unique T_B-geodesic joining x to y. The preceding Bass–Serre-tree observation gives $xC = yC$, so $x, y \in uC$ for some $u \in B$.

These endpoints are intermediate vertices of q, so $u \notin C$; such a subpath cannot begin or end at an endpoint of q, because then its first or last edge would be an a- or b-edge in H, which is impossible. Since r is a subpath of a geodesic and B is isometrically embedded on vertices, each replacement preserves length. The resulting path is a T_B-geodesic from 1 to v^k, hence is the segment in L. Each nontrivial replacement segment lies in both L and uL and has length at least 4, because its distinct endpoints belong to uC. This contradicts (8). Thus q is the segment in the adjacent B-tree between its endpoints.

Successive vertices of C on L are at distance 4, so every point of this segment is within distance 2 of H. It follows, by decomposing any Γ-geodesic with endpoints in H into its maximal subpaths whose endpoints lie in H and whose intermediate vertices lie outside H, together with the parts lying in H, that the whole geodesic is within distance 2 of the vertex set H. Since H has diameter one in Γ_H, its diameter in Γ_H is at most $2+1+2=5$. Figure 3(a) illustrates such a genus-four subpath and the added edge in the base coset.

In both cases, left translation gives a uniform bound for every Γ-geodesic joining the endpoints of an added H-edge. For the original generator edges the required bound is immediate. Proposition 2.2 proves that Γ_H is hyperbolic.

Condition (c): the relative metric. Let q be an admissible path from 1 to $h \in H$. Its image in $\mathcal{T}$ avoids the vertex H, since precisely the forbidden H-edges in the base coset would pass through that vertex. The midpoints of C and hC must therefore lie in the same component of $\mathcal{T} \setminus \{H\}$. Hence $hC = C$, or $h \in C$. Thus

$$\hat{d}(1,h) = \infty \quad (h \in H \setminus C).$$

Take a shortest admissible path from 1 to v^k. Such a path exists, since the word v^k in $X^{\pm 1}$ is admissible. It cannot contain a maximal subpath whose endpoints x, y lie in B and whose intermediate vertices lie outside B. Indeed, the preceding Bass–Serre-tree observation gives $xC = yC$. The first edge of such a subpath has an H-label, so admissibility implies $x \notin H$. If $x \neq y$, the single H-edge joining x to y lies in $xH \neq H$ and is allowed, giving a shorter path. If $x = y$, delete the subpath. This is again a contradiction. Thus a shortest admissible path stays in B.

It may still use H-edges in other cosets; these must not be discarded. Every such edge has endpoints in uC for some $u \in B \setminus C$, because $B \cap H = C$. By (9) their projections to L are at distance at most one; see Figure 3(b). The same bound holds for the endpoints of each $X^{\pm 1}$-edge, since projection onto a subtree is 1-Lipschitz. Projecting the path edge by edge therefore gives

$$2m|k| = d_{T_B}(1, v^k) \leq \ell(q).$$

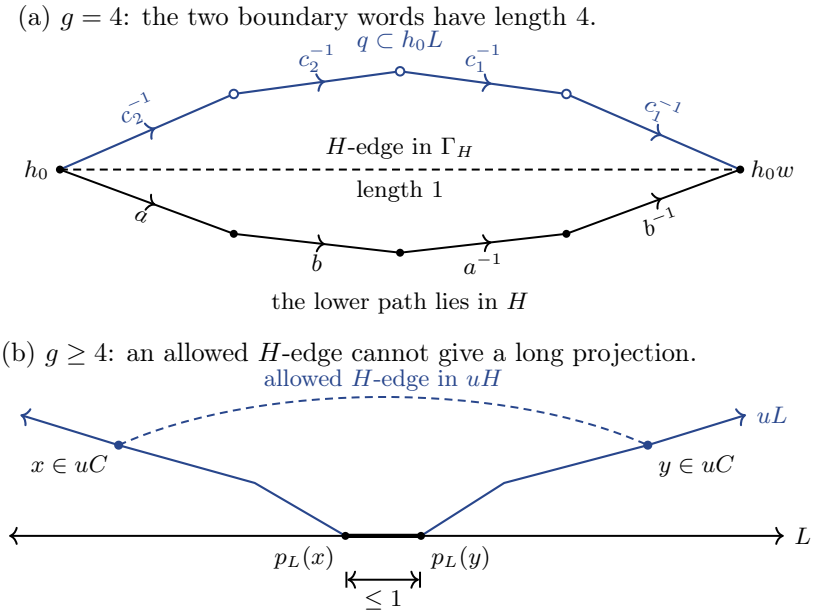


FIGURE 3. The estimates in Proposition 4.3. (a) Here $h_0 \in H$ and $w = [a, b] = c_2^{-2} c_1^{-2}$. Every point of the upper path is within distance 2 of H, whose diameter in Γ_H is one. The dashed edge lies in the forbidden base coset H. (b) For $u \in B \setminus C$, an allowed H-edge with endpoints in uC projects to a segment of L of length at most one. The axes are drawn schematically, not to scale.

The X-word v^k realizes equality. Since $w = v^{-1}$, this proves (7). In particular, for every $r \geq 0$ the closed relative ball is

$$\overline{B}_{\hat{d}}(1, r) = \{w^k : k \in \mathbb{Z},\ |k| \leq \lfloor r/(2m) \rfloor\}. \tag{12}$$

Left translation by H preserves admissibility, so every finite-radius ball is finite. This proves condition (c) and completes the proof. □

Remark 4.4 (Alternative proof using Bowditch's criterion)**.** Here is a second proof of Proposition 4.3, without the relative-metric computation. We will show that H is a proper infinite quasiconvex malnormal subgroup of the non-elementary hyperbolic group G_g. Bowditch's theorem [2, Theorem 7.11], applied to the conjugacy class of H, then shows that G_g is hyperbolic relative to H. By [9, Proposition 4.28], this is equivalent to hyperbolic embeddedness with respect to any finite relative generating set, in particular $X = \{c_1, \ldots, c_m\}$.

We check the hypotheses. For $m \geq 2$, the cyclic word $v = c_1 c_1 \cdots c_m c_m$ has no proper period: the cyclic two-letter word $c_1 c_2$ occurs exactly once. Hence v is not a proper power, and C is a maximal cyclic subgroup of the free

group B, therefore malnormal in B. For completeness, if $C \cap uCu^{-1}$ contains a nontrivial element, the axes of powers of v and of uvu^{-1} coincide. The stabilizer of this axis in a free group is generated by the primitive root of v, which here is v itself. Thus $u \in C$; this is also a standard consequence of the free action on the Cayley tree [26, Chapter I].

The subgroup H is then malnormal in $H *_C B$. Indeed, for $x \notin H$, an element of $H \cap xHx^{-1}$ fixes the segment joining the distinct vertices H and xH of the Bass–Serre tree. This segment contains a B-vertex with two distinct incident edges. Their stabilizers are, up to conjugacy, C and uCu^{-1} with $u \in B \setminus C$, and their intersection is trivial.

Finally, H is quasiconvex. For $g \geq 4$, the curve $\delta = \partial T$ bounds neither a disk nor a Möbius band, so it can be represented by a simple closed geodesic in a hyperbolic metric on N_g. After an isotopy, T therefore has geodesic boundary. A component $\widetilde{T}$ of its preimage in $\mathbb{H}^2$ is an intersection of hyperbolic half-planes, hence convex. Its stabilizer is H, and $\widetilde{T}/H = T$ is compact. Thus an H-orbit is coarsely dense in $\widetilde{T}$ and quasiconvex in $\mathbb{H}^2$. The Milnor–Švarc lemma and stability of quasi-geodesics transfer this to a Cayley graph of G_g; see [6, Chapters I.8 and III.H]. The subgroup is infinite and proper by the amalgam normal form. Bowditch's criterion now applies.

The consequence for bounded cohomology follows from the following extension result.

Theorem 4.5 (Frigerio–Pozzetti–Sisto, [12, Proposition 6.1])**.** *If $H \hookrightarrow_h G$, restriction $EH_b^n(G) \to EH_b^n(H)$ is surjective for every $n \geq 2$. Consequently, restriction is also surjective on reduced exact bounded cohomology.*

Indeed, restriction is norm non-increasing, so it sends zero-seminorm classes to zero-seminorm classes. Hence the surjection on exact bounded cohomology descends to a surjection on the reduced exact quotients.

Corollary 4.6. For the figure-eight embedding i and $g \geq 4$, the map $i^n : \overline{EH}_b^n(G_g) \to \overline{EH}_b^n(F_2)$ is surjective for every $n \geq 2$.

4.4. **Genus three and isometric extension in bounded cohomology.** When $g = 3$, we have $m = 1$, so the complementary surface $N_{m,1}$ is a Möbius band and

$$G_3 = \langle a, b, c \mid [a, b]c^2 = 1 \rangle = F(a, b) *_{\langle [a,b] \rangle = \langle c^{-2} \rangle} \langle c \rangle. \tag{13}$$

The edge group remains infinite cyclic. Its inclusion into the Möbius-band vertex group is $\mathbb{Z} \to \mathbb{Z}$, $k \mapsto -2k$; its image has index two, not order two.

Proposition 4.7. The free subgroup $H = \langle a, b \rangle$ in (13) is not hyperbolically embedded in G_3 with respect to any relative generating set.

Proof. The orientation character $G_3 \to \{\pm 1\}$ is trivial on H and sends c to -1, so $c \notin H$. But $c^2 = [a,b]^{-1} \in H$, and

$$\langle c^2 \rangle \subseteq H \cap cHc^{-1}.$$

This intersection is infinite, so Proposition 2.4 applies. Directly, in $\Gamma(G_3, \{c\} \sqcup H)$ the path

$$1 \xrightarrow{c} c \xrightarrow{c^{2k} \in H} c^{2k+1} \xrightarrow{c^{-1}} c^{2k}$$

is admissible for every $k \geq 1$: its middle edge lies in cH. Hence $\hat{d}(1, c^{2k}) \leq 3$ and the relative metric is not proper. This is exactly the shortcut that the axis-overlap estimate prevents when $g \geq 4$. □

Remark 4.8. In fact, no π_1-injective embedded figure-eight in N_3 with orientable regular neighborhood has almost malnormal image. To see this, let R be that regular neighborhood. Since $\chi(R) = -1$, it is either a once-holed torus or a pair of pants. The complement has Euler characteristic zero. It has no disk component, since a disk would kill a nontrivial boundary element of $\pi_1(R)$. Each component has boundary and nonpositive Euler characteristic; consequently each is an annulus or a Möbius band. There are either one or three boundary components in total, so at least one complementary component is a Möbius band. Its core, based using a path in R, gives an orientation-reversing element x outside $\pi_1(R)$ with $x^2 \in \pi_1(R) \setminus \{1\}$. Thus $\langle x^2 \rangle \subseteq \pi_1(R) \cap x\pi_1(R)x^{-1}$ is infinite. The obstruction is therefore geometric, not merely a poor choice of the generators a, b.

The obstruction does not prevent extension of bounded cohomology. We use the following specialization of the graph-of-groups theorem of Bucher, Burger, Frigerio, Iozzi, Pagliantini and Pozzetti.

Theorem 4.9 ([8, Theorem 1.1])**.** *Let A, B be countable groups, let D be amenable, and suppose the maps $D \to A$ and $D \to B$ are injective. Put $G = A *_D B$, and let $\iota_A : A \hookrightarrow G$ and $\iota_B : B \hookrightarrow G$ be the canonical inclusions. For $n \geq 2$, write*

$$\mathrm{res}_A = \iota_A^* : H_b^n(G) \longrightarrow H_b^n(A),$$
$$\mathrm{res}_B = \iota_B^* : H_b^n(G) \longrightarrow H_b^n(B)$$

for the induced restriction maps. There is a linear map

$$\Theta^n : H_b^n(A) \oplus H_b^n(B) \longrightarrow H_b^n(G)$$

such that, for every $u \in H_b^n(A)$ and $v \in H_b^n(B)$,

$$\mathrm{res}_A\big(\Theta^n(u,v)\big) = u, \qquad \mathrm{res}_B\big(\Theta^n(u,v)\big) = v,$$
$$\|\Theta^n(u,v)\| = \max\{\|u\|, \|v\|\}.$$

Thus Θ^n is an isometric linear embedding for the maximum seminorm on the direct sum. The first two identities say that it is a right inverse to the simultaneous restriction map

$$H_b^n(G) \longrightarrow H_b^n(A) \oplus H_b^n(B), \qquad \xi \longmapsto \big(\mathrm{res}_A(\xi), \mathrm{res}_B(\xi)\big).$$

No malnormality or maximality assumption on the images of D occurs in this theorem. In particular, the index-two boundary inclusion of the Möbius band is allowed.

Proposition 4.10. For every $g \geq 3$ and $n \geq 2$, restriction for the figure-eight embedding admits an isometric linear right inverse

$$J_g^n : H_b^n(F_2) \longrightarrow H_b^n(G_g), \qquad i^* J_g^n = \mathrm{Id}\,.$$

It descends to an isometric linear map

$$\overline{J}_g^n : \overline{EH}_b^n(F_2) \longrightarrow \overline{EH}_b^n(G_g), \qquad i^n \overline{J}_g^n = \mathrm{Id}\,. \tag{14}$$

Thus the required restriction i^n is surjective also for $g = 3$.

Proof. In (5), the two vertex groups $H = F_2$ and $B = F_m$ are countable. The edge group $C_0 \cong \mathbb{Z}$ is amenable, and both edge maps are injective by (6). Apply Theorem 4.9 to these edge maps, with $A = H = F_2$ and $G = G_g$. The restriction res_A in that theorem is precisely i^*. Define

$$J_g^n(u) = \Theta^n(u, 0).$$

Then $i^* J_g^n(u) = u$. Moreover,

$$\|u\| = \|i^* J_g^n(u)\| \leq \|J_g^n(u)\| \leq \|(u, 0)\| = \|u\|,$$

so J_g^n is isometric. When $g = 3$, $B = \mathbb{Z}$ and $H_b^n(B) = 0$ for $n \geq 1$, but the argument needs no modification.

We check exactness before passing to reduced cohomology. The surface N_g is aspherical for $g \geq 3$, so it is a $K(G_g, 1)$. Since N_g is closed and non-orientable, $H^2(N_g; \mathbb{R}) = 0$, and its cohomology in degrees above two vanishes by dimension. Hence

$$H^n(G_g; \mathbb{R}) = 0 \qquad (n \geq 2).$$

Also $H^n(F_2; \mathbb{R}) = 0$ for $n \geq 2$, since a graph is a $K(F_2, 1)$. These are the standard surface and graph cohomology calculations; see [16, Sections 1.B and 3.1]. Therefore

$$EH_b^n(G_g) = H_b^n(G_g), \qquad EH_b^n(F_2) = H_b^n(F_2) \quad (n \geq 2).$$

Isometry shows that J_g^n sends zero-seminorm classes to zero-seminorm classes, so it descends to the map in (14). If η has zero seminorm, then $\|\xi + \eta\| = \|\xi\|$ by the triangle inequality in both directions; hence the quotient norm is the seminorm of any representative. It follows that the descended map is again isometric. The identity with restriction also descends, proving surjectivity. □

Remark 4.11. The maps J_g^n and $\overline{J}_g^n$ are cohomological sections. They are not asserted to be pullbacks of a group or surface retraction. In particular, Proposition 4.10 does not assert a hyperbolic embedding in genus three. It supplies exactly the restriction surjectivity needed below, and gives an alternative to Corollary 4.6 in every genus $g \geq 4$ as well.

4.5. **Construction of the annular pushing homomorphisms.** Fix the figure-eight embedding i from Section 4.2. Its two loops have annular neighborhoods inside the orientable subsurface T. Thus the construction of [4, Lemma 3.1] applies locally, regardless of the orientability of N_g. We provide the details, keeping the cochain conventions of Section 3.

Lemma 4.12. *For every $g \geq 3$ there exist $\Lambda > 0$ and homomorphisms $\rho_\epsilon : F_2 \to \mathcal{H}_g$, $0 < \epsilon < 1/2$, such that, for every $n \geq 2$ and $\xi \in \overline{EH}_b^n(G_g)$,*

$$\left\| \rho_\epsilon^n \overline{E\Gamma}_b^n(\xi) - \Lambda i^n(\xi) \right\| \longrightarrow 0 \qquad (\epsilon \to 0). \tag{15}$$

The constant Λ and the family ρ_ϵ are independent of n and ξ.

Proof. Let α, β be the two loops in the image of the figure-eight embedding i. Choose closed smooth annular neighborhoods $N(\alpha), N(\beta) \subset \mathrm{int}(T)$ whose intersection is a disk containing z in its interior. Put

$$\Lambda = \mu(N(\alpha) \cap N(\beta)) > 0.$$

On either annulus the measure μ is represented by a smooth positive area form after choosing an orientation.

Use the model annulus

$$N = \{(r, \theta) : 1 \leq r \leq 2,\ \theta \in \mathbb{R}/2\pi\mathbb{Z}\}, \qquad \kappa = r\, dr \wedge d\theta.$$

There is an orientation-preserving diffeomorphism $n_\alpha : N(\alpha) \to N$ for which

$$n_\alpha^*(s_\alpha \kappa) = \nu_\alpha, \qquad s_\alpha = \frac{\mu(N(\alpha))}{\int_N \kappa} > 0,$$

where ν_α represents μ on $N(\alpha)$. Indeed, start with an orientation-preserving parametrization and apply Moser's argument after this normalization of total area; the version for manifolds with boundary is due to Banyaga [22, 1]. The parametrization can be chosen so that increasing θ represents the chosen orientation of α. Choose n_β similarly. Notice that equal areas of the two annuli, or equality with the Euclidean area of N, are not required.

Choose a smooth function $f_\epsilon : [1, 2] \to [0, 1]$ that equals 1 on $[1 + \epsilon, 2 - \epsilon]$ and vanishes in neighborhoods of 1 and 2. The isotopy

$$P_\epsilon^t(r, \theta) = (r, \theta + 2\pi t f_\epsilon(r)), \qquad 0 \leq t \leq 1,$$

preserves κ and every constant multiple of it. Conjugate by n_α, respectively n_β, and extend by the identity outside the corresponding annulus. This gives measure-preserving isotopies $P_\epsilon^t(\alpha), P_\epsilon^t(\beta)$ of N_g. Identifying F_2 with $F(a, b)$, define the homomorphism $\rho_\epsilon : F_2 \to \mathcal{H}_g$ by

$$\rho_\epsilon(a) = P_\epsilon^1(\alpha), \qquad \rho_\epsilon(b) = P_\epsilon^1(\beta).$$

Freeness ensures that these assignments define a homomorphism; injectivity of ρ_ϵ is not needed.

Put

$$A_\epsilon(\alpha) = n_\alpha^{-1}\{1+\epsilon \le r \le 2-\epsilon\}, \qquad B_\epsilon(\alpha) = N(\alpha) \setminus A_\epsilon(\alpha),$$

and define the analogous sets for β. The time-one map $\rho_\epsilon(a)$ fixes $A_\epsilon(\alpha)$ pointwise, although the isotopy traces one full loop around α there. Consequently, for $x \in A_\epsilon(\alpha)$, $\gamma(\rho_\epsilon(a), x)$ is conjugate to $i_*(a)$; for $x \notin N(\alpha)$ it is trivial. The analogous statements hold for b; see Figure 4.

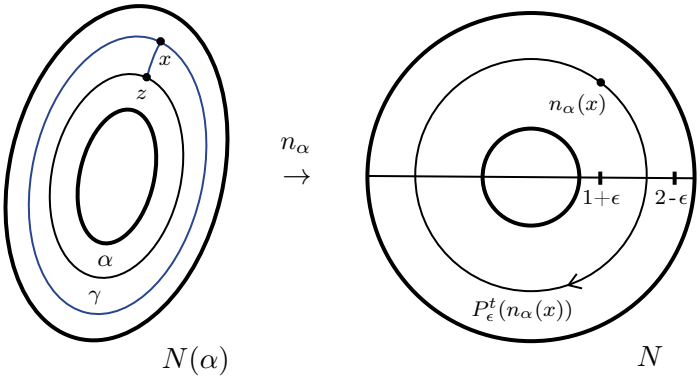


FIGURE 4. The annular pushing isotopy. For $x \in A_\epsilon(\alpha)$, the time-one map fixes x and the trajectory class $\gamma(\rho_\epsilon(a), x)$ is conjugate to $i_*(a)$.

Up to boundary sets of measure zero, decompose N_g into

$$A_\epsilon = A_\epsilon(\alpha) \cap A_\epsilon(\beta), \qquad A_\epsilon^a = A_\epsilon(\alpha) \setminus N(\beta),$$
$$A_\epsilon^b = A_\epsilon(\beta) \setminus N(\alpha), \qquad B_\epsilon = B_\epsilon(\alpha) \cup B_\epsilon(\beta),$$
$$D = N_g \setminus (N(\alpha) \cup N(\beta)).$$

The five sets displayed above form a measurable partition up to null boundary sets. Both time-one generators fix $N_g \setminus B_\epsilon$ pointwise; since they are bijections, B_ϵ is invariant as well. Thus every piece of the partition is $\rho_\epsilon(F_2)$-invariant. Let $h_a : F_2 \to \langle a \rangle$ send a to a and b to 1, and define h_b similarly. Because the intersection $N(\alpha) \cap N(\beta)$ is a disk, the conjugating paths for the two trajectory loops can be chosen simultaneously there. Thus, for $\omega \in F_2$, Proposition 3.2 gives

$$\gamma(\rho_\epsilon(\omega), x) = \begin{cases} u_x i_*(\omega) u_x^{-1}, & x \in A_\epsilon, \\ u_{x,a} i_* h_a(\omega) u_{x,a}^{-1}, & x \in A_\epsilon^a, \\ u_{x,b} i_* h_b(\omega) u_{x,b}^{-1}, & x \in A_\epsilon^b, \\ 1, & x \in D, \end{cases} \tag{16}$$

where the conjugators depend on x, but not on ω. For example, on A_ϵ choose, measurably in x, a path from z to x inside the intersection disk and compare it with the reference path $\alpha_{z,x}$. On A_ϵ^a and A_ϵ^b use paths inside the

corresponding annulus. The resulting path classes, and hence the conjugators, are measurable and take values in the countable group G_g. Since these points are fixed by $\rho_\epsilon(F_2)$, (2) and the cocycle identity imply

$$\eta(\rho_\epsilon(\omega), x) = \gamma(\rho_\epsilon(\omega), x) \qquad (x \notin B_\epsilon). \tag{17}$$

No explicit formula is needed on B_ϵ.

Choose a bounded homogeneous cocycle c such that $\zeta = [c] \in EH_b^n(G_g)$ maps to $\xi \in \overline{EH}_b^n(G_g)$ under the reduced quotient. For $\bar\omega = (\omega_0, \dots, \omega_n)$, the cochain representing the pullback of the transfer is

$$\bar\omega \longmapsto \int_{N_g} c\big(\eta(\rho_\epsilon(\omega_0), x), \dots, \eta(\rho_\epsilon(\omega_n), x)\big)\, d\mu(x).$$

The integral over each of the invariant pieces above is a homogeneous cocycle, by the proof of Proposition 3.3 applied to that piece.

Let $c_{\epsilon,A} \in C_b^n(F_2)^{F_2}$ denote the contribution of A_ϵ to this cocycle. Equations (16) and (17) give

$$c_{\epsilon,A}(\omega_0, \dots, \omega_n) = \int_{A_\epsilon} c\big(u_x i_*(\omega_0) u_x^{-1}, \dots, u_x i_*(\omega_n) u_x^{-1}\big)\, d\mu(x).$$

We prove that $c_{\epsilon,A}$ is cohomologous, through a bounded primitive, to the cocycle $\mu(A_\epsilon) i^* c$. This also justifies the use of varying conjugators. For $u \in G_g$ let $R_{u^{-1}}$ denote right multiplication by u^{-1} on vertices, and define the prism operator

$$(K_u^n c)(g_0, \dots, g_{n-1}) = \sum_{j=0}^{n-1} (-1)^j c(g_0, \dots, g_j, g_j u^{-1}, \dots, g_{n-1} u^{-1}).$$

It preserves left homogeneity, has operator norm at most n, independently of u, and satisfies

$$\delta K_u^n c + K_u^{n+1} \delta c = R_{u^{-1}}^* c - c.$$

The identity follows by cancellation in the alternating sum; it is the usual prism homotopy (cf. [16, Section 2.1]). By left invariance of c, pullback by conjugation by u equals $R_{u^{-1}}^* c$. Apply this identity with $u = u_x$, pull back by i_*, and integrate over A_ϵ. More explicitly, the pointwise integral

$$b_{\epsilon,A} = \int_{A_\epsilon} i^*\big(K_{u_x}^n c\big)\, d\mu(x) \in C_b^{n-1}(F_2)^{F_2}$$

is well-defined: the integrand is measurable and $\|b_{\epsilon,A}\|_\infty \le n\mu(A_\epsilon)\|c\|_\infty$. Since $\delta c = 0$, the homotopy identity gives

$$c_{\epsilon,A} - \mu(A_\epsilon) i^* c = \delta b_{\epsilon,A}.$$

Consequently the two bounded-cohomology classes are equal:

$$[c_{\epsilon,A}] = \mu(A_\epsilon)[i^* c] = \mu(A_\epsilon) i^*(\zeta) \quad \text{in } H_b^n(F_2) = EH_b^n(F_2).$$

Here exactness follows from $n \geq 2$ and $H^n(F_2; \mathbb{R}) = 0$. Writing $\overline{[c_{\epsilon,A}]}$ for the image of $[c_{\epsilon,A}]$ in the reduced quotient, the corresponding equality is

$$\overline{[c_{\epsilon,A}]} = \mu(A_\epsilon) i^n(\xi) \quad \text{in } \overline{EH}^n_b(F_2). \tag{18}$$

The same argument on A^a_ϵ and A^b_ϵ reduces their contributions to pullbacks through h_a and h_b. Both factor through an infinite cyclic group, whose positive-degree bounded cohomology is zero. Thus both contributions vanish in cohomology. The contribution of D also vanishes, since it factors through the trivial group. These amenable-group vanishing statements are recalled in [13].

Let R_ϵ be the cocycle obtained by integrating over B_ϵ. It satisfies

$$\|R_\epsilon\|_\infty \leq \mu(B_\epsilon)\|c\|_\infty.$$

The preceding computations give

$$\rho_\epsilon^* \Gamma^n_b(\zeta) = \mu(A_\epsilon) i^*(\zeta) + [R_\epsilon] \quad \text{in } H^n_b(F_2).$$

As above, $EH^n_b(F_2) = H^n_b(F_2)$ for $n \geq 2$, so the error class $[R_\epsilon]$ is exact. Denote its image in the reduced quotient by $\overline{[R_\epsilon]}$. Passing to that quotient and using (18), we obtain

$$\rho_\epsilon^n \overline{E\Gamma}^n_b(\xi) = \mu(A_\epsilon) i^n(\xi) + \overline{[R_\epsilon]} \quad \text{in } \overline{EH}^n_b(F_2).$$

The transition collars have measure tending to zero, and

$$\mu(B_\epsilon) \longrightarrow 0, \qquad \mu(A_\epsilon) \longrightarrow \mu(N(\alpha) \cap N(\beta)) = \Lambda.$$

Consequently,

$$\begin{aligned} \left\| \rho_\epsilon^n \overline{E\Gamma}^n_b(\xi) - \Lambda i^n(\xi) \right\| &\leq |\mu(A_\epsilon) - \Lambda| \, \|i^n(\xi)\| + \mu(B_\epsilon)\|c\|_\infty \\ &\longrightarrow 0. \end{aligned}$$

This proves the lemma. □

4.6. **Completion of the proof.**

Proof of Theorem 1. Fix $g \geq 3$ and $n \geq 2$. By Proposition 4.10, the map $\overline{J}^n_g$ is an isometric linear section of i^n. We claim that

$$\mathcal{F}^n_g = \overline{E\Gamma}^n_b \circ \overline{J}^n_g : \overline{EH}^n_b(F_2) \longrightarrow \overline{EH}^n_b(\mathcal{H}_g)$$

is injective. If $\mathcal{F}^n_g(u) = 0$, apply Lemma 4.12 to $\xi = \overline{J}^n_g(u)$. Since $i^n(\xi) = u$, it gives

$$\|\Lambda u\| = \left\| \rho_\epsilon^n \mathcal{F}^n_g(u) - \Lambda u \right\| \longrightarrow 0.$$

The left side is independent of ϵ, and $\Lambda > 0$, so $u = 0$ in the reduced space. This proves injectivity and hence

$$\dim H^n_b(\mathcal{H}_g) \geq \dim \overline{EH}^n_b(\mathcal{H}_g) \geq \dim \overline{EH}^n_b(F_2).$$

Thus, more generally, for every $n \geq 2$ we have constructed an injection

$$\overline{EH}^n_b(F_2) \hookrightarrow \overline{EH}^n_b(\mathcal{H}_g).$$

In degrees two and three the source has dimension $2^{\aleph_0}$, by the references given at the beginning of this section. In particular, both bounded cohomology groups in the theorem are infinite-dimensional. □

Remark 4.13. Surjectivity alone is sufficient for the dimension argument. Indeed, Lemma 4.12 implies

$$\ker(\overline{E\Gamma}_b^n) \subseteq \ker(i^n).$$

Thus Corollary 4.6 also proves the result for $g \geq 4$ without using the section $\overline{J}_g^n$. For the elementary retraction in Section 4.1, the directions are j^n surjective and r^n injective, with $j^n r^n = \mathrm{Id}$. The uniform section in Proposition 4.10 avoids separate final arguments for the three genus ranges.

Remark 4.14. An interesting problem is to compute $H_b^n(\mathrm{Homeo}_0(N_g, \mu))$ for $n \geq 2$ and $g = 1, 2$. New ideas are required for this method: $\pi_1(N_1)$ is finite and $\pi_1(N_2)$ is virtually abelian, so both are amenable and have trivial bounded cohomology in positive degrees.

Remark 4.15. It was recently proved that $H_b^4(F_2)$ is nontrivial; see [18]. Since $H^4(F_2; \mathbb{R}) = 0$, one has $H_b^4(F_2) = EH_b^4(F_2)$. What is not currently known is whether a nonzero class survives after quotient by the zero-seminorm classes, that is, whether $\overline{EH}_b^4(F_2)$ is nontrivial. Any such nonzero reduced class would, by the argument above, give a nonzero class in $H_b^4(\mathrm{Homeo}_0(N_g, \mu))$ for every $g \geq 3$.

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Department of Mathematics, Ben Gurion University, Israel

Email address: brandens@bgu.ac.il

Department of Mathematics, Ben Gurion University, Israel

Email address: lior.me10@gmail.com